

Scaling intra-urban climate fluctuations

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Urban-induced microclimate variations, such as urban heat islands and air pollution, scale with city size, producing distinctive relations between average climate variables and city-scale quantities (for example, total population). However, these relations are sensitive to city boundary definitions and overlook intra-urban variability. Here we overcome these limitations by using high-resolution data of urban temperatures, air quality, population and street networks from 142 cities worldwide, showing that their marginal and joint probability distributions collapse onto a set of general functions inspired by finite-size scaling in statistical physics. Through a logarithmic relation linking urban spatial features to climate variables, we find that average street network properties are sufficient to characterize the full variability of temperature and air pollution fields within and across cities. These findings show that intra-urban climate variability follows general scaling functions, enabling the integration of climate information into reduced-complexity models of urban systems to better inform future urban planning.

High temperature and air pollution in cities increase the risk of heat- and pollution-related mortality, posing serious economic and public health challenges^{1–3}. In parallel, urban areas are growing, with projections indicating that 68% of the global population will reside in cities by 2050⁴. Understanding how urbanization impacts local climate is therefore critical to reduce risks and shape healthier urban environments^{5,6}.

While the physical processes behind urban-induced changes—such as the urban heat island (UHI) effect and the increase in urban particulate matter (PM) concentrations—are well understood, existing research has mostly focused on city-specific studies based on detailed monitoring and modeling efforts (for example, ref. 7). This has substantially advanced our understanding of mass and energy transfers in urban contexts, including the impacts of different building materials, urban geometries and human activities on local microclimate (for example, ref. 7), but the identification of general patterns and laws that are valid across diverse cities and geographic contexts is lagging behind^{8–10}. Urban complexity research has emerged as a powerful approach to explore such emergent behaviors^{11–13}, providing the methods and theoretical frameworks that are needed to describe the structure and dynamics of cities in the most general terms. Yet, translating this knowledge into a general description of urban microclimate remains an open challenge.

Consider an urban area characterized by some properties x (for example, population) that are heterogeneous in space, that is $x = x(s)$,

where s is the spatial coordinate (Fig. 1a). Such urban characteristics modify local climate variables y (for example, temperature) by perturbing the background climatic conditions $\langle y_0 \rangle$, where the brackets $\langle \rangle$ indicate spatial averaging. This induces a distinct field of urban–rural differences $\Delta y(s) = y(s) - \langle y_0 \rangle$, which are often aggregated at the city level to define quantities such as the UHI intensity (for example, ref. 14), that is, $\langle \Delta y \rangle = \frac{1}{A} \int_A y(s) ds - \langle y_0 \rangle$, where A is the urban area. Drawing parallels to allometric scaling in biological systems (for example, refs. 15,16), previous studies have applied scaling laws to quantify how these aggregate climate variables scale with city size, that is, $\langle \Delta y \rangle \propto X^b$, where the climate perturbation $\langle \Delta y \rangle$ is defined in terms of air temperature^{17,18}, surface temperature^{14,19} or air pollution^{20,21}, b is the scaling exponent and $X = \int_A x(s) ds$ is a given city-scale urban feature generally defined by total urban population, urban area or gross building volume (for example, refs. 14,17).

However, unlike biological systems, cities lack clear perimeters, and the construction of scaling laws depends on the choice of the area A , thus inevitably introducing biases related to boundary definitions. Previous studies^{22,23} have shown that even small variations in density thresholds, commuting flows or population cutoffs used to delineate urban areas lead to different scaling exponents b . Beyond these parametric approaches, the choice between administrative versus metropolitan boundaries can also yield opposite conclusions—for example, on whether larger cities emit more or less CO₂ per capita

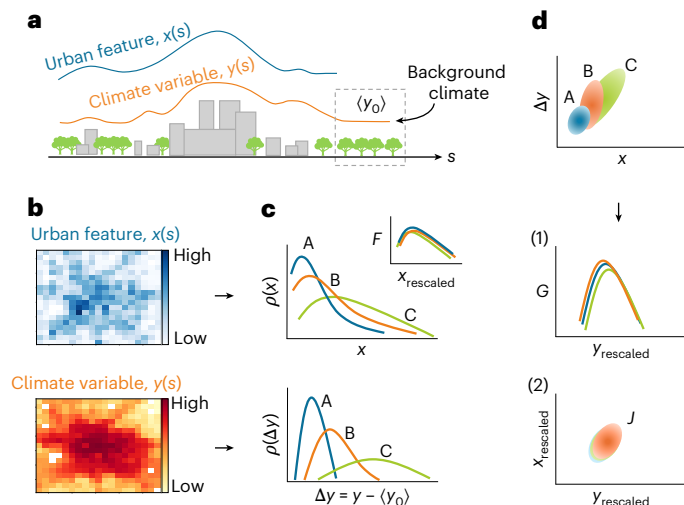


Fig. 1 | Conceptual framework for analyzing the covariation of urban structure and climate. **a**, In this study, we consider intra-urban variations of climate variables $y(s)$ (that is, temperature T and PM concentrations) and their relation to urban features $x(s)$ (that is, street intersections n and population counts p), where s is the spatial coordinate. Urban–rural differences $\Delta y = y - \langle y_0 \rangle$ are calculated considering background climatic conditions $\langle y_0 \rangle$. **b**, An example of a regular square grid used to aggregate all datasets, allowing both climate variables and urban features to be compared across cities. **c**, PDFs of urban features $\rho(x)$ and urban climate perturbations $\rho(\Delta y)$, where each curve represents one city. Previous work²⁸ showed that the PDFs of urban features collapse onto a single curve following an FFS form F (inset). **d**, The methodology employed in this study: first we investigate the relation between x and Δy and then we rescale their marginal and joint PDFs to obtain a data collapse—indicating the existence of general scaling functions (G and J , see main text).

(that is, $b > 1$ versus $b < 1$)²⁴. Moreover, urban systems can be highly heterogeneous in terms of infrastructure and population densities (for example, ref. 25), with local climatic conditions varying from building to neighborhood scales owing to diverse materials, green spaces, urban geometries and anthropogenic activities (for example, refs. 26,27).

With regards to urban features x , these limitations have been recently overcome by Hendrick et al.²⁸, who—inspired by analogous findings on biological metabolism^{29–32}—demonstrated that the entire variability of population and street intersections within a city, as described by their probability density functions (PDFs) $\rho(x)$, follows a finite-size scaling (FSS) form

$$\rho(x|\langle x \rangle) = x^{-\lambda} F\left(\frac{x}{\langle x \rangle^\delta}\right), \quad (1)$$

where F is a scaling function providing a finite-size cutoff, and λ and δ are the scaling exponents that describe how the distributions change with system size and provide a physical interpretation of the characteristic scale. The two exponents are not independent owing to mathematical constraints (that is, normalization). When $\delta = 1$, the distributions have the same shape and differ only in their characteristic size, whereas for $\delta \neq 1$, their shape varies systematically. Therefore, the value of the scaling exponent determines the physical meaning of the scale parameter (that is, $\langle x \rangle^\delta$ in equation (1)). In this framework, since $\lambda \approx \delta \approx 1$, the scale corresponds to the mean value $\langle x \rangle$, which is therefore sufficient to characterize the full probability distribution of urban features $\rho(x)$ —that is, using equation (1), distributions from cities around the world collapse onto a single master curve F , which is approximately log-normal²⁸. More broadly, this approach is rooted in the FSS framework of statistical physics^{33,34}, where ‘scaling laws’ observed in finite systems are understood as effective manifestations of an underlying scale-invariant behavior that

is regularized by finite size. In FSS, one does not expect pure power laws to hold unconditionally; instead, system-size dependence is absorbed into a characteristic scale, and the residual variability is captured by a universal scaling function—revealed empirically through data collapse onto a master curve. Equation (1) is precisely of this form: the exponent λ controls the apparent scaling regime, while $\langle x \rangle^\delta$ has the role of the size-dependent cutoff that encodes how the finite urban ‘system’ constrains fluctuations. Since cities are finite in size, interpreting them through the lens of FSS naturally reframes allometric relations as ‘effective’ scaling behaviors and provides a principled route to compare heterogeneous cities by separating universal structures (the master curve) from size-, morphology- and climate-dependent corrections to scaling.

The FSS framework serves here as a point of departure to explore whether a similar ansatz also exists for urban climate (see Fig. 1 for an overview of our conceptual framework and methodology). As cities differ widely in size, morphology and background climate, their temperature and pollution distributions are not directly comparable. Thus, we investigate whether the probability distributions of climate variables $\rho(y)$ follow a general scaling form. In addition, given that aggregate urban climate statistics are generally unknown for cities around the world, we ask whether information on urban structure x is sufficient to characterize the entire intra-urban variability of climate (defined here as mean annual temperature and air pollution fields).

The description of the spatial variability of climatic conditions has long been a central focus in urban climatology (for example, ref. 7). Yet, a general statistical framework describing intra-urban climate variations—and its link to urban form and characteristics—is still missing. An attempt in this direction is represented by radial decay models^{35,36}, which relate urban-induced microclimate changes to the distance from the city center. These approaches indicate that both temperature and pollution levels follow a decreasing trend with radial distance, but they focus only on the mean behavior of climatic characteristics, thus neglecting the spatial heterogeneities of cities—which can markedly deviate from average coarse-grained quantities. Here, we overcome all the abovementioned limitations by showing that marginal probability distributions of urban climate—and their joint distributions with urban features—are characterized by general scaling functions that, upon proper rescaling, collapse onto a unique curve for each climate variable, reflecting a common statistical form shared across cities. These results provide a novel stochastic description of intra-urban climate variability that is valid across cities, generalizes previous findings on urban structure²⁸ and improves traditional urban climate decay models (for example, ref. 35). Given the climate challenges faced by cities worldwide, this knowledge is critical to assess the environmental exposure of urban residents and inform climate-sensitive urban planning strategies.

Results

Our analysis investigates intra-urban climate variability using high-resolution datasets from 142 cities worldwide—ranging from small towns with tens of thousands of inhabitants (for example, Tartu, Estonia) to megacities with populations exceeding 1 million (for example, Bogotá, Colombia), spanning all major climate zones and levels of urban development across the world (Fig. 2c and Supplementary Table 1). Specifically, we analyze near-surface air temperature (T) and PM concentrations ($\text{PM}_{2.5}$, denoted as PM for simplicity)—which represent the generic climate variable y —alongside the number of street intersections (n) and population counts (p)—which represent the urban features x (see the Methods for details).

To ensure consistent spatial resolution across cities, we discretize space into a regular grid of square cells of size $l = 1,000$ m, aggregating all the data within each cell (Fig. 1b and Methods). Thus, we obtain pairs (x_i, y_i) for each grid cell i , where x_i is the urban feature (for example, street intersections) and y_i is the climate variable (for example, temperature). For urban climate, we define urban and nonurban grid

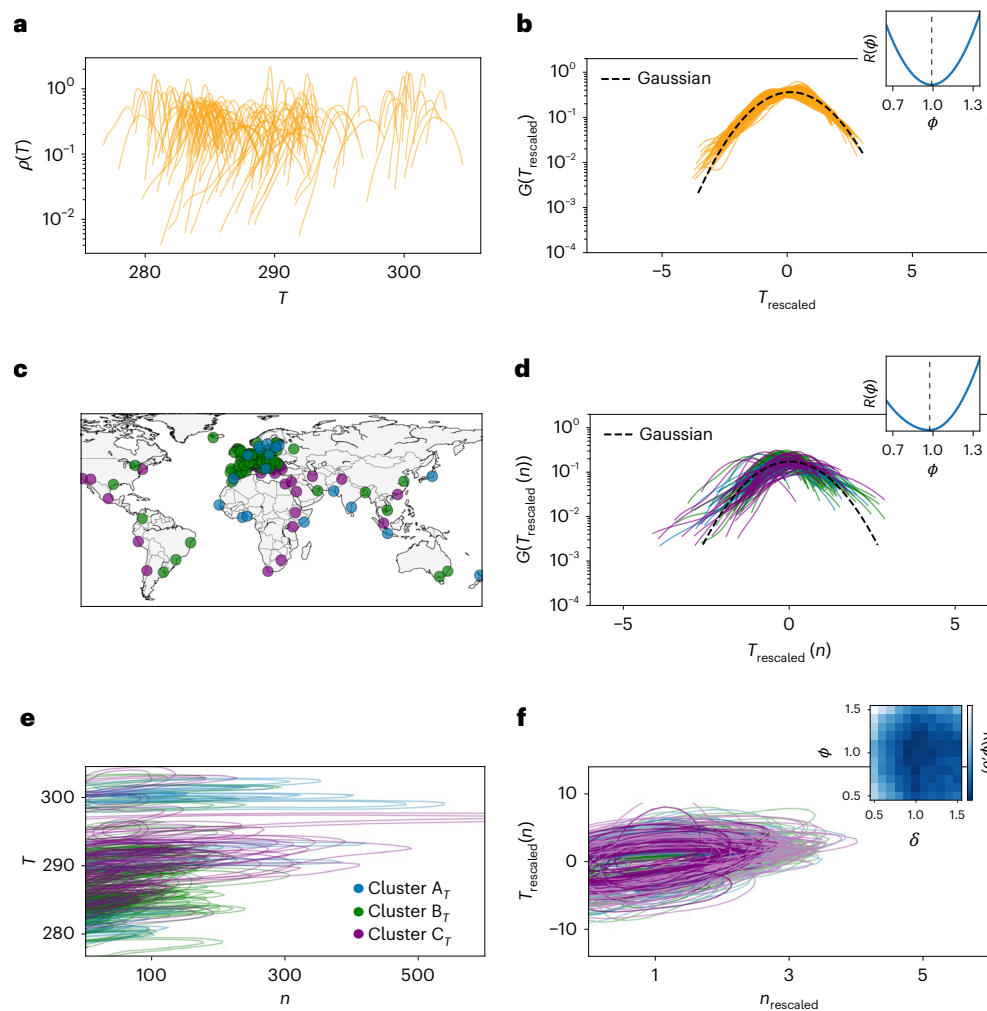


Fig. 2 | Scaling intra-urban temperature variations and covariations with street network intersections. **a**, Distinct PDFs of temperature observed across the analyzed cities, where each curve represents one city. **b**, After rescaling, using the empirical climate data according to equations (2) and (3), these PDFs collapse onto a common scaling function G . A Gaussian fit (dashed black line) indicates that G is well approximated by a normal distribution. The inset shows the residual $R(\phi)$. **c**, The global distribution of cities and their grouping into three clusters. **d**, Collapse of the marginal PDFs after rescaling, using the street network

information according to equations (6) and (7). As for **b**, the Gaussian fit (dashed black line) and residual $R(\phi)$ (inset) are also shown. **e**, Joint PDFs for temperature (T) and street network intersections (n) for all cities analyzed. The colors refer to the different clusters illustrated in **c**. **f**, After both variables are rescaled using equation (8), the data collapse follows a common joint scaling function. The inset shows the residual $R(\phi, \delta)$. Street network information from OpenStreetMap (<https://www.openstreetmap.org>).

cells and calculate urban–rural differences for each urban grid cell i as $\Delta y_i = y_i - \langle y_0 \rangle$, where $\langle y_0 \rangle$ is computed as the average value of the non-urban grid cells surrounding each city (see the Methods for details).

Scaling intra-urban climate variations across cities

While distributions of urban features are strongly asymmetric and can be fully characterized by the mean²⁸, climate distributions appear shifted and nearly symmetric around the mean (see Figs. 2a and 3a for temperature and pollution, respectively). Hence, for climate variables, we propose a location-scale ansatz that differs from equation (1) as it requires two rather than one parameter—the location $\langle \Delta y \rangle$ and the scale parameter $\sigma_{\Delta y}^y$. As cities differ in their background climate and in how strongly urban form perturbs that field, we (1) remove each city's rural baseline to isolate the urban contribution and (2) locate and scale each PDF by a characteristic amplitude of intra-urban climate intensity and fluctuations. In particular, we found that the PDFs of the climate variables follow a scaling ansatz of the form

$$\rho(y | \langle \Delta y \rangle, \sigma_{\Delta y}) = \frac{1}{\sigma_{\Delta y}^\phi} G(y_{\text{rescaled}}), \quad (2)$$

where G is a general scaling function that holds across all cities. The rescaled climate variable is defined as

$$y_{\text{rescaled}} = \frac{y - \langle y_0 \rangle - \langle \Delta y \rangle}{\sigma_{\Delta y}^\phi}, \quad (3)$$

where ϕ and γ are the scaling exponents and $\langle y_0 \rangle$ is the rural mean of temperature or PM concentrations (that is, $\langle T_0 \rangle$ and $\langle \text{PM}_0 \rangle$, respectively). This approach allows collapsing the different PDFs into a single curve G (both for T and PM), revealing that the intra-urban climate fields can be described by a general statistical form across cities (Figs. 2b and 3b).

Linking climate perturbations to urban features

The result in equation (2), while notable for its validity across all the analyzed cities, relies on a priori knowledge of urban climate variables (that is, $\langle \Delta y \rangle$ and $\sigma_{\Delta y}$), which are generally unknown. Hence, as a next step, we seek to predict the climate PDFs in equation (2) using information on urban features rather than empirical statistics of T and PM. To do so, we propose a simple model that describes their

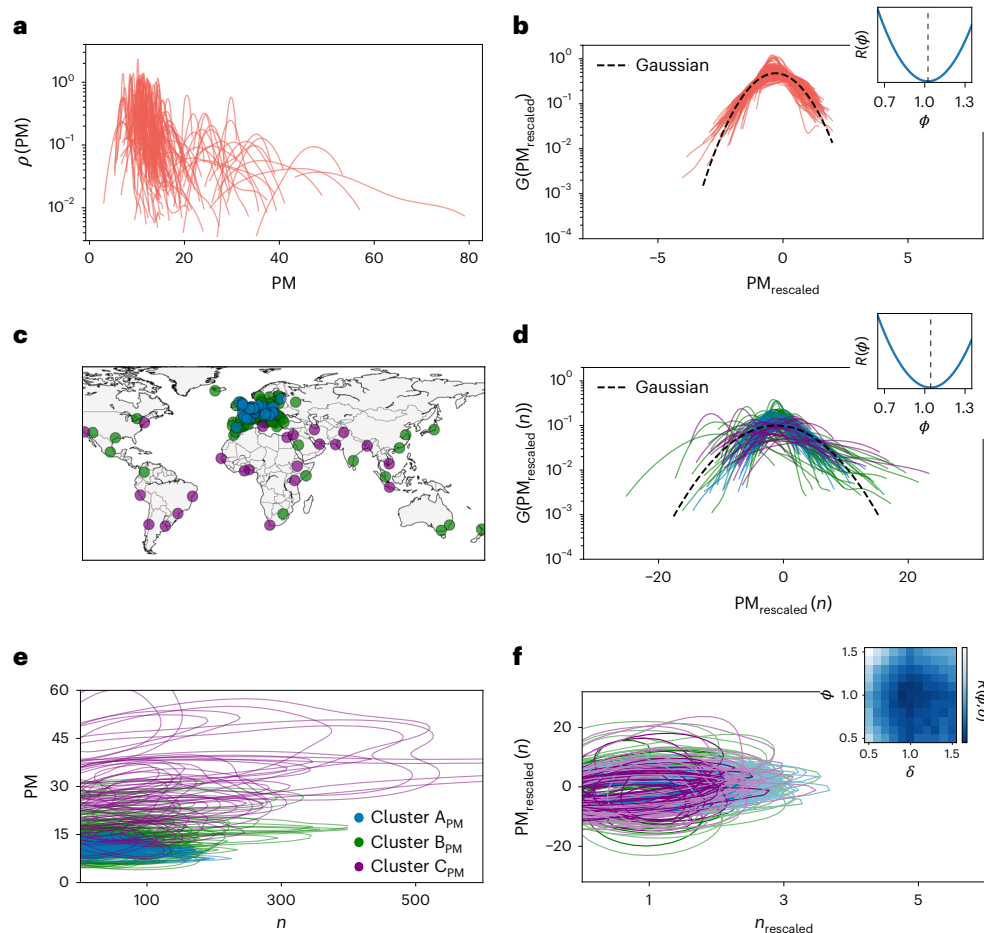


Fig. 3 | Scaling intra-urban air quality (PM) variations and covariations with street network intersections (n). **a**, Distinct PDFs of PM concentrations observed across the analyzed cities, where each curve represents one city. **b**, After rescaling, using the empirical climate data according to equations (2) and (3), these PDFs collapse onto a common scaling function G . A Gaussian fit (dashed black line) indicates that G is well approximated by a normal distribution. The inset shows the residual $R(\phi)$. **c**, The global distribution of cities and their grouping into three clusters. **d**, Collapse of the marginal PDFs

after rescaling using the street network information according to equations (6) and (7). As for **b**, the Gaussian fit (dashed black line) and residual $R(\phi)$ (inset) are also shown. **e**, Joint PDFs for temperature and street network intersections for all cities analyzed. The colors refer to the different clusters illustrated in **c**. **f**, After both variables are rescaled using equation (8), the data collapse follows a common joint scaling function. The inset shows the residual $R(\phi, \delta)$. Street network information from OpenStreetMap (<https://www.openstreetmap.org>).

covariation within cities. Specifically, we found that urban–rural differences in temperature ΔT (that is, the magnitude of UHI) and PM concentration ΔPM are well captured by a logarithmic relationship with either population counts or the number of street intersections (Supplementary Fig. 1). Explicitly, for the grid cell i in a given city, this relationship is expressed as

$$\Delta y_i = \alpha + \beta \ln x_i, \quad (4)$$

where α and β are city-dependent intercept and slope, respectively. The analysis of the α and β values across all the cities analyzed reveals that their magnitude is more constrained when derived from street intersections compared to population counts (Supplementary Fig. 1). Note that equation (4) is statistically appropriate because T and PM are approximately normally distributed (Figs. 2a and 3a), while urban characteristics follow approximately log-normal distributions²⁸. For the sake of completeness, we also performed a numerical comparison with linear and power-law models and discussed a possible multivariate extension of the logarithmic model (for further details, see Supplementary Table 2 and Supplementary Information, section 6). However, our primary goal is not to predict fine-scale climate values in each square cell i (as typically performed by regression models;

for example, ref. 37), but rather to predict aggregate statistical properties of climate field—that is, the location and scale. Thus, applying the expected value and variance to both sides of equation (4) yields the following expressions for the mean and s.d. of climate variables

$$\langle \Delta y(x) \rangle = \alpha + \beta \langle \ln x \rangle, \quad \sigma_{\Delta y(x)} = \beta \sigma_{\ln x}, \quad (5)$$

which provide a link between quantifiable urban feature information and generally unknown urban climate variables. Thanks to the relations in equation (5), we can now estimate the location and scale parameter in equation (2) using the statistical properties of urban features and reconstruct the distributions of urban climate variables without any a priori information on their statistics.

To reduce the number of free parameters while retaining generality, we replace city-specific intercepts α and slopes β with a small set of cluster-specific averages, $\bar{\alpha}_k$ and $\bar{\beta}_k$, where $k \in \{1, \dots, K\}$ indexes clusters and K is the total number of clusters. Intuitively, clustering allows cities that have similar environmental contexts (regional climate, local emissions regulations or topography) to share parameters, yielding a parsimonious model with $2K$ parameters instead of $2 \times$ (number of cities). Thus, with cluster-specific parameters, equation (2) becomes

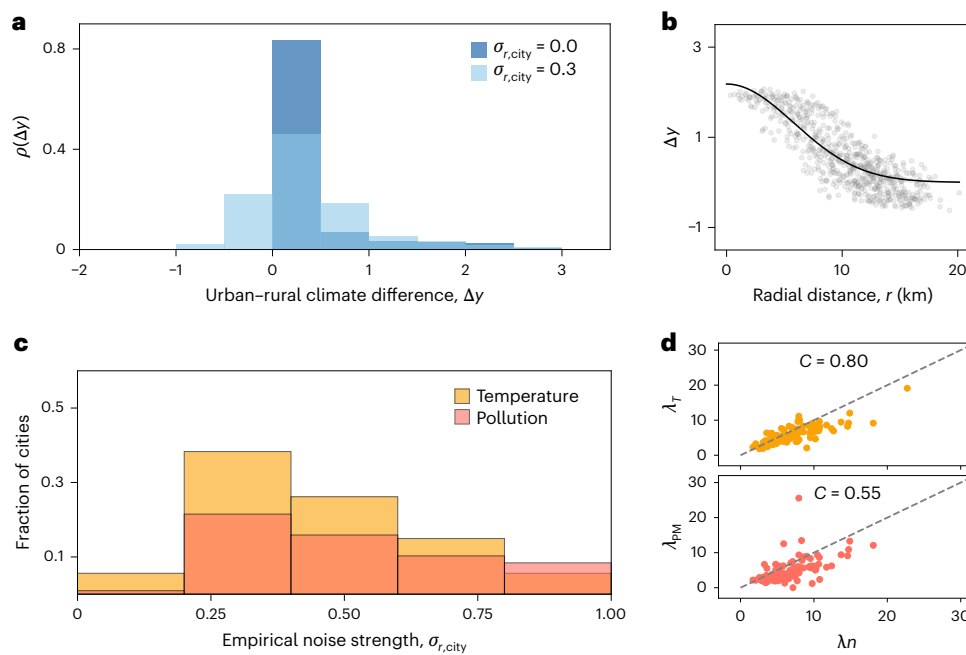


Fig. 4 | Stochastic radial decay model of climate variables. **a**, The PDFs of Δy from numerical simulations: when $\sigma_{r,\text{city}} = 0.0$, the deterministic decay model is recovered, while for $\sigma_{r,\text{city}} = 0.3$, an approximately Gaussian PDF shape emerges, using a peak value $y_A = 2.5$, decay rate $\lambda_y = 5$ km and city radius $R = 25$ km. **b**, An illustration of the radial decay of climate variables (that is, urban–rural temperature or pollution differences, Δy) together with the deterministic decay

model fit for Bucharest. **c**, The distribution of empirical noise strengths ($\sigma_{r,\text{city}}$) of climate variables across all cities. **d**, Scatter plots of the street network decay rate (λ_n) versus both the decay rate of temperature (λ_T) and PM (λ_{PM}), along with their corresponding Pearson correlation values C , evaluated using two-sided tests, with exact P values of $P = 4.3 \times 10^{-25}$ for temperature and $P = 4.9 \times 10^{-9}$ for PM.

$$\rho(y | \langle \ln x \rangle, \sigma_{\ln x}) \times (\bar{\beta}_k \sigma_{\ln x})^\gamma = G(y_{\text{rescaled}}(x)), \quad (6)$$

where the rescaled climate variable is now defined as

$$y_{\text{rescaled}}(x) = \frac{y - \langle y_0 \rangle - (\bar{\alpha}_k + \bar{\beta}_k \langle \ln x \rangle)}{(\bar{\beta}_k \sigma_{\ln x})^\phi}, \quad (7)$$

where x denotes the chosen urban feature (street intersections n or population counts p).

We first tested a single pair of global parameters ($K = 1$), that is, intercept $\bar{\alpha}$ and slope $\bar{\beta}$ common to all 142 cities. The resulting data collapse for both temperature and PM revealed distinct families of curves (Supplementary Fig. 3), indicating that environmental context matters in addition to urban features, which we quantified using the k -means clustering method to group cities with shared parameter values $\bar{\alpha}_k$ and $\bar{\beta}_k$ (see the Methods for details). While both street intersections and population can serve as predictors for rescaling the urban climate PDFs, Supplementary Fig. 2 shows that, for temperature, fewer clusters K are needed when using street intersections, whereas for pollution, both predictors perform similarly. Consequently, we adopt the street network ($x = n$) as our predictor, for which the optimal partition is $K = 3$ (see Fig. 2c for temperature and Fig. 3c for PM and see Supplementary Table 3 and Supplementary Information, section 3 for details on cluster characteristics).

After rescaling via equations (6) and (7), all PDFs collapse onto a common curve (Figs. 2d and 3d), indicating that, once structural differences are accounted for, intra-urban climate variability follows a common statistical form—consistent with evidence that local anomalies are modulated by urban morphology and form^{17,18}. For details on the rationale behind the clustering effect on the data collapse, we refer readers to Supplementary Information, section 3.

To identify the scaling exponents that yield the best data collapse, we used a residual metric R ³⁸, which measures the cumulative area

enclosed between all pairs of rescaled PDF curves. This provides a quantitative criterion for the quality of the data collapse. A lower R indicates better collapse, reflecting that the rescaled curves are closer together. The scaling exponents are not independent owing to the normalization constraint $\int_0^\infty \rho(y) dy = 1$ ($\phi = \gamma$; Supplementary Information, section 1). Thus, minimizing R over ϕ shows that the best collapse occurs for $\phi = \gamma \approx 1$ (insets in Figs. 2d and 3d). This result—consistent with the proportionality of moments derivations—indicates that the distributions preserve their shape across cities and differ only in their characteristic scale, which corresponds to the standard deviation—confirming that intra-urban climate distributions are fully characterized by their mean and variance (Supplementary Fig. 5 and Supplementary Information, section 1).

Note that the functional forms of the collapsed curves are reasonably well approximated by a normal distribution (Figs. 2b–d and 3b–d, dashed black line). As a matter of fact, equation (2) can be derived using the results by Hendrick et al.²⁸ for urban features—that is, equation (1)—together with the logarithmic relation in equation (4), leading to a Gaussian form of G (see derivation in Supplementary Information, section 2).

Covariation of intra-urban characteristics and climate variables

After showing that the PDFs of temperature and PM concentrations follow the scaling ansatz in equation (2), we now investigate whether similar arguments apply to their joint probability distributions with urban features (Figs. 2e and 3e). Since urban climate variability is influenced by multiple factors, it is crucial to examine the coupled fluctuations of urban characteristics and climate, and verify whether their covariation also follows a general scaling function²⁸. To this end, we test whether the joint PDFs $\rho(x, y)$ satisfy the following functional form

$$\rho(x, y | \langle x \rangle, \langle \Delta y \rangle, \sigma_{\Delta y}) = \frac{1}{x^\lambda} \frac{1}{\sigma_{\Delta y}^\gamma} J\left(\frac{x}{\langle x \rangle^\delta}, \frac{y - \langle y_0 \rangle - \langle \Delta y \rangle}{\sigma_{\Delta y}^\phi}\right), \quad (8)$$

where f is a general scaling function that holds across all cities, and ϕ , γ , λ and δ are the scaling exponents. As before, we use equation (5) along with the same clusters identified for the marginal PDFs using the street network ($x = n$; $K = 3$; see Fig. 2c for temperature and Fig. 3c for PM), and rewrite equation (8) as a function of urban features

$$\rho(x, y | \langle x \rangle, \langle \ln x \rangle, \sigma_{\ln x}) x^{\lambda} (\beta_k \sigma_{\ln x})^{\gamma} = J(x_{\text{rescaled}}, y_{\text{rescaled}}(x)), \quad (9)$$

where $y_{\text{rescaled}}(x)$ is defined as in equation (7), and the rescaled urban feature is defined as in ref. 28

$$x_{\text{rescaled}} = \frac{x}{\langle x \rangle^{\delta}}. \quad (10)$$

After this rescaling, Figs. 2f and 3f show that the joint probability distributions of climate with street intersections, $\rho(T, n)$ and $\rho(\text{PM}, n)$, achieve a robust collapse onto a single function (visualized in two dimensions as contours of a surface), indicating that once structural differences are accounted for, intra-urban characteristics and climate covariations also follow a common statistical form.

To identify the scaling exponents that yield the best data collapse, we again employ the residual metric R^{38} , minimizing R over the two independent scaling exponents ϕ and δ , which, under the normalization constraint, results in $\phi = \gamma$ for the climate variable y (Supplementary Information, section 1), and $\lambda = 1$ for the urban feature x (see ref. 29). The insets in Figs. 2f and 3f show that the best data collapse occurs at $\phi \approx \delta \approx 1$. This result—consistent with the proportionality of moments derivations—indicates that the joint distributions preserve their shape across cities and differ only in their characteristic scale—corresponding to the standard deviation for climate variables, and the mean for urban features (for example, street intersections)—consistently with the results from ref. 28 and our previous findings (Supplementary Fig. 5 and Supplementary Information, section 1).

Reconciling spatial decay models with observed probability distributions

Our results reveal that, when appropriately rescaled by urban structure, the probability distributions of climate variables collapse onto an approximately Gaussian form (Figs. 2 and 3). Yet, traditional radial decay models—which relate urban-induced microclimate changes to the distance from the city center^{35,36}—yield PDFs that deviate from the observed Gaussian behavior (even when polycentric cities are disregarded; Methods), following instead the form $\rho(y) \propto 1/y$ (see Fig. 4a and Supplementary Information, section 4 for the mathematical derivations). Here, we show that this discrepancy arises from the deterministic formulation, which overlooks the substantial fluctuations around the mean decay trend observed in empirical data (Fig. 4b). To show this, we extend the traditional deterministic decay function of urban–rural climate differences by adding an additive stochastic term, that is,

$$\Delta y(r) = y_A \exp\left(-\frac{r^2}{2\lambda_y^2}\right) + \mathcal{N}(0, \sigma_{r,\text{city}}^2), \quad (11)$$

where r denotes the radial distance from the city center (see the Methods for definitions), y_A is the climate central value at $r = 0$, λ_y controls the decay rate and $\mathcal{N}(0, \sigma_{r,\text{city}}^2)$ represents white noise with standard deviation $\sigma_{r,\text{city}}$ —which prompts the PDFs to shift from non-Gaussian to Gaussian shapes (Fig. 4a). To quantify the strength of the stochastic component $\sigma_{r,\text{city}}$ in the data, we compute it as the standard deviation of the residuals—defined as the difference between the observed empirical values and the fitted deterministic decay profile (equation (11))—for each city's temperature and PM concentration fields (see the Methods for details). Figure 4c shows the resulting distribution of city-level $\sigma_{r,\text{city}}$ values across all cities analyzed.

We further explored whether the decay of urban climate is linked to urban characteristics. To do so, we applied the exponential decay

model in equation (11) to both street intersections and population counts (Supplementary Fig. 10), this time incorporating multiplicative white noise, which reproduces the observed log-normal PDFs of urban features²⁸ (see Supplementary Fig. 11 and Supplementary Information, section 4 for the mathematical derivations). Consistent with our previous results, the street network again emerges as a more reliable descriptor of climate variability than population counts, showing stronger and more significant correlations between decay rates and stochastic noise strengths across cities (Supplementary Table 4). Notably, cities with stronger street intersection decay rates λ_n also tend to exhibit stronger urban climate decay rates for both temperature λ_T and pollution λ_{PM} (Fig. 4d). However, the Pearson correlations between the noise component of street intersections and that of climate variables are moderate for temperature and negligible for pollution (Supplementary Table 4), suggesting that additional processes (for example, industrial emissions or atmospheric transport) have an important role.

In summary, these findings provide a conceptual bridge between traditional radial decay studies and our scaling framework, highlighting the key role of urban structure in shaping intra-urban climate variability.

Discussion

We have shown that marginal and joint probability distributions of intra-urban air temperature and PM concentrations across 142 cities, when rescaled by characteristic scales derived from the urban structure, collapse into general scaling functions that approximate a Gaussian form (Figs. 2 and 3). While inspired by FFS ideas from statistical physics^{33,34}, the observed collapse reflects a location-scale statistical regularity of intra-urban climate fields that deviates from the traditional FSS ansatz, with size-dependence replaced by the variance of the climatic variable (equation (2)). As such, this study advances existing scaling approaches based on city-level metrics^{14,17–21} by extending them to the entire intra-urban variability of climate, and further shows that climatic variables and their underlying urban structures follow different statistical forms²⁸. As a result, we overcome long-standing problems related to the definition of city boundaries^{22,23} and the neglect of local urban heterogeneities (for example, ref. 25), which would otherwise mask important microclimatic effects^{26,27}. This framework does not aim to model specific physical or chemical processes but rather to uncover generic spatial statistical regularities that emerge across cities with diverse morphologies, climatic conditions and cultural contexts. Similarly, recent work has shown that urban–rural differences in the temporal occurrence of heat events across cities can be predicted using a standardized Gaussian framework that combines the mean UHI with rural temperature variance and persistence³⁹. This parallel suggests that both temporal and spatial urban climate fluctuations can be described within a common statistical paradigm.

While the general scaling holds across all cities, clustering reveals intercity differences beyond urban morphology—such as regional climate, local emissions regulations or topography (see Supplementary Table 3 and Supplementary Information, section 3 for details). In particular, rescaling the climate fields on the basis of street network information led to $K = 3$ clusters (Figs. 2c and 3c) that improve the data collapse but do not change the underlying shape function G , which is shared across all cities.

The methodological robustness of our approach is confirmed by several sensitivity tests. Specifically, we demonstrate scale independence by recalculating our results using square grid cells of different sizes (see the Methods and Supplementary Fig. 4), and confirm the general validity by repeating the analysis with independent datasets for both temperature and pollution (see Supplementary Figs. 6 and 7 and Supplementary Information, section 5 for details).

We further show that traditional radial decay models for urban climate^{35,36} produce heavy-tailed PDFs that deviate from the near-Gaussian behavior observed here. By adding an additive stochastic term to the radial decay model—empirically derived and quantified

from the data—we are able to reproduce the observed distributions. This reconciliation suggests that intra-urban climate is shaped by the interplay between the predictable decay with distance of infrastructure density—explained, for example, by gravitational models (for example, ref. 40)—and the randomness inherent in heterogeneous urban morphologies.

In our study, we used two simple urban predictors—street intersections and population counts—because they are globally available and consistently defined and reflect built form and human activities, which are key drivers of urban heat and pollution. More detailed covariates (for example, vegetation or land cover) may improve local predictability where available but would reduce the generality of global cross-city scaling analyses⁴¹. Notably, we find that the street network is a more reliable predictor of intra-urban climate variability than population counts. This is reflected in the more consistent fitted parameters (α , β) in equation (4) (Supplementary Fig. 1), the tighter data collapse achieved with fewer clusters K (Supplementary Fig. 2) and the stronger correlations between decay rates and stochastic noise strengths across cities (Supplementary Table 4). From a physical perspective, this is explained by the fact that urban climates depend on the geometry and energy budgets of different urban surfaces, which are directly linked to infrastructure characteristics—while population is only an indirect measure of urban form.

Our results are consistent with the findings by La Porta et al.⁴², who demonstrated that the probability distributions of total CO₂ and PM across different population ranges share a common shape, once normalized by each city's total population. Thus, while their work reveals general fluctuations at the city scale, our results extend their insight to the intra-urban level, uncovering common behaviors driven by heterogeneous spatial structures, within and across cities. In the context of intra-city variations, Shreevastava et al.^{43,44} investigated the patterns of land surface temperature and identified urban heat islets—that is, clusters of high surface temperatures—characterized by power-law size distributions. Our results complement their findings by demonstrating that the entire variability of urban climate characteristics—defined across uniform spatial units rather than heat islets—exhibits general scaling patterns.

This work has several limitations that open up clear next steps. First, we used annual means to ensure comparability across cities worldwide and to emphasize persistent structural differences in intra-urban climate, while minimizing the influence of transient events such as heatwave episodes. However, annual mean temperature and PM fields overlook temporal dynamics, and further research could incorporate changes on diurnal^{45,46} and seasonal^{47,48} timescales, thereby enabling a full characterization of the corresponding PDFs over time. Moreover, PM concentrations show a slightly weaker collapse than temperature, which is consistent with the fact that air pollution is influenced by advection and localized emission sources—factors that go beyond urban morphology. A key challenge is therefore to include richer descriptors without losing the generality and transferability of our results.

In conclusion, this study provides a statistical framework, inspired by FSS theory, that fully describes intra-city fluctuations of urban microclimate and links them to simple measures of urban form. The simplicity of the approach makes it particularly valuable for cities with limited computational capacity, as well as for incorporating climate information into reduced-complexity models of urban systems, from transport to economic and energy systems⁴⁹. This opens up the opportunity to (1) benchmark urban climate models by testing their ability to reproduce the observed scaling of marginal and joint PDFs, (2) downscale regional simulations to the intra-urban level and (3) obtain first-order estimates of spatial patterns of urban climate in cities where detailed monitoring/modeling is unavailable. Our framework further advances the theoretical understanding of how urban form shapes intra-urban climate dynamics and establishes a direct bridge

between urban complexity theories and the quantitative description of urban climate variability. Given the urgency of climate mitigation and adaptation, our results suggest that cities worldwide should speed up the uptake of zero-emission technologies as urban structure, mobility and pollution are inherently coupled. In addition, considering the observed covariation of population with the intensity of urban climate perturbations, there is a pressing need for strategies that balance the positive effects of higher urban densities (fostering, for example, active mobility⁵⁰) with the increased exposure of urban residents to climate-related risks¹.

Methods

Data sources

This study includes 142 cities for which urban climate simulations⁵¹ were available. Although the sample mostly contains European cities—reflecting the coverage of the urban climate database—it spans all major climate zones and levels of urban development across the world, providing a diverse global sample. The selection was constrained solely by the availability of simulated air temperature data, without applying additional filters. All cities included had sufficient spatial extent to ensure meaningful intra-urban comparisons.

We used mean daily ambient temperature estimates at 2 m above ground with a 100 m × 100 m spatial resolution produced by the UrbClim model⁵¹ via the VITO UrbClim FTP server (Data Availability and Code Availability). UrbClim was developed by the Flemish Institute for Technological Research, and it integrates a land surface scheme with a three-dimensional atmospheric boundary layer module, incorporating land cover, soil sealing, vegetation and meteorological data from the fifth-generation European Centre for Medium-Range Weather Forecasts reanalysis. For this study, annual averages of temperature estimates were computed for the entire year 2017.

The annual average PM_{2.5} concentrations are derived from a machine-learning-based global dataset⁵². The dataset provides daily PM_{2.5} estimates at a 1-km resolution for the period 2017–2022, generated using a model that integrates multiple data sources, including satellite aerosol optical depth retrievals (for example, MODIS MAIAC and GEOS-FP), ground-based air-quality monitoring stations (9,500 globally), meteorological variables (for example, temperature, humidity and wind speed), emissions inventories and land use information. A spatiotemporally enhanced machine learning model (4D-STET) is used to fill gaps in aerosol optical depth observations and to account for complex relationships between pollution, meteorology and surface features. For this study, we use the annual average for 2022 to characterize PM exposure across cities.

The street networks are extracted from OpenStreetMap (OSM)⁵³, an open-access, collaborative mapping project that provides continuously updated, editable maps of the world. The data are processed using OSMnx⁵⁴ to analyze the structural variability of urban transport networks, offering detailed information on streets, intersections, buildings and other infrastructure. In this study, we specifically use this dataset to analyze street network intersections. While OSM provides the most comprehensive global database of urban street networks, its completeness can vary across cities—particularly in historical cores. However, our analysis focuses on statistical properties aggregated over thousands of grid cells per city, which mitigates the impact of local incompleteness.

Finally, we use global population count data from the WorldPop dataset⁵⁵, which integrates remote sensing, census data and geospatial technologies to produce high-resolution population distribution maps. These datasets provide consistent and comparable population estimates across urban and rural areas, with resolutions as fine as 100 m × 100 m in some regions. For this study, we use the unconstrained global mosaic population dataset at 1-km resolution for the year 2020, which was created using a top-down disaggregation approach based on statistical models and satellite-derived covariates. The dataset is

designed to provide consistent 1-km resolution global population estimates, making it suitable for large-scale comparative urban studies.

To validate the robustness of our findings, we compare the results with alternative datasets. For PM_{2.5} concentration, we use a dataset covering Europe downloaded from the European Environmental Agency for the year 2022 (ref. 56), and for temperature we use a dataset of global land surface temperature for the year 2013 (ref. 57) (see Supplementary Information, section 5 for details).

Data integration

To ensure consistent spatial resolution across cities, we aggregate all data into a regular square grid. Each square cell has a side length size $l = 1,000$ m (Fig. 1b), partitioning the urban area into spatially uniform units across all cities for the analysis, thereby defining urban features as density measures. Within each cell, we compute the mean of all temperature and PM_{2.5} measurements, and the sum of all street intersections and population counts. A sensitivity analysis with different side length sizes l is provided in Supplementary Fig. 4. The cells of the different scales have been delimited to keep spatial coherence: four 500-m cells form one of the 1-km cells used in the previous figures, and four 1-km cells aggregate to form a 2-km cell. Note that population and PM_{2.5} data are available only at 1-km resolution; therefore, the 500-m were performed only for temperature and street network data.

We define a distinction between urban and rural areas using the street network. Specifically, we consider as urban grid cells where there is at least one intersection ($n \geq 1$), and rural where there is only one intersection ($n = 1$) to avoid water zones. The average rural climate value $\langle y_0 \rangle$ is computed at a 1,000-m resolution—where the agreement with the UrbClim classification of rural areas⁵¹, based on land use and built-environment characteristics, is highest. This urban–rural distinction is used to remove the average rural climate signal and to isolate the mean and scale of urban climate in a consistent manner across all spatial resolutions l for the ansatz (equation (2)), which reflects the city's perturbation of the background climate field. This procedure ensures that the rural climate baseline remains physically consistent across scales, while allowing the urban variability to be assessed at multiple spatial resolutions.

We adopt the spatial domain provided by our UrbClim temperature dataset⁵¹ for all cities. While previous research suggests that city structure follows consistent radial profiles that scale with population size⁵⁸, enabling rescaled urban domains based on population-dependent functions, we find that applying such rescaling does not alter our results. In fact, we tested multiple variants of the rescaled radial extent and observed nearly identical data collapses, as long as the same criterion was applied consistently across all cities.

Statistical analysis

We use kernel density estimation⁵⁹ because it provides a flexible and nonparametric approach to estimate and derive PDFs for urban climate and urban characteristics variables. PDFs are computed for population counts, street network intersections, air temperature and PM concentrations, providing insights into the spatial distribution and variability of these variables within cities. In addition, the quality of the data collapse—both for marginal and joint PDFs—is quantitatively assessed using the residual-based metric proposed in ref. 38, which evaluates the overall distance between rescaled curves.

Clustering

We use the K -means clustering method⁶⁰ because it provides a simple and effective way to cluster cities on the basis of similarity in their data collapse response, while requiring no prior assumptions about cluster shape or distribution. For each city, we first computed the rescaled climate distribution using a global intercept $\bar{\alpha}$ and slope $\bar{\beta}$ values averaged across all 142 cities ($K = 1$; Supplementary Fig. 3)—and summarize its form by the coordinates of its maximum density point, $(x_{\text{peak}}, y_{\text{peak}})$.

Thus, each city is represented by a two-component feature vector. We run K -means for $k = 1, \dots, 10$, which partitions cities by minimizing the within-cluster sum of squared Euclidean distances in the two-dimensional $(x_{\text{peak}}, y_{\text{peak}})$ space. The optimal number of clusters is selected using the elbow of the within-cluster sum of squares, expressed as explained variance. Using the street network led to $K = 3$ for both temperature and pollution (Supplementary Fig. 2). Finally, to ensure the tightest possible collapse of the rescaled curves, we recomputed the Bhattacharjee et al. residual metric R^{38} and found that only 4.2% for temperature (6 cities) and 9.9% for pollution (14 cities) changed cluster assignment, confirming the robustness of our grouping and very low computational cost.

Radial decay model

We defined city center as the geographic centroid of all the street intersection coordinates, computed as the mean of their latitude and longitude values (independently of the regular square grid system). For each grid cell centroid (longitude, latitude) within the gridding system, the radial distance r was calculated as the haversine (that is, geodesic) distance between its centroid coordinates and the city center ($r = 0$). This geodesic formulation ensures that distances are physically consistent across cities located at different latitudes and longitudes, providing a reproducible reference framework for the radial decay analysis. To ensure the validity of the radial decay framework—which assumes approximate circular symmetry—we excluded cities with poor fits for the street intersection decay model ($R^2 < 0.1$), keeping 107 out of 142 cities (76%) for this spatial decay analysis. An additional 12 cities were excluded from the PM analyses owing to implausible decay rates.

To test the white-noise assumption in the stochastic climate radial decay model, we divided each city into concentric radial belts, each containing the same number of data points. We then calculated the standard deviation of the decay-model residuals (σ_r) within each belt. This procedure allowed us to test whether residual fluctuations were approximately uniform across distance (Supplementary Fig. 8). Supplementary Figs. 9 and 10 provide fit diagnostics—residual histograms and R^2 statistics—for both climate variables and urban features.

Note that the selection of a subset of cities for the decay model analysis does not alter the generality of the scaling functions describing the marginal and joint PDFs—which are the main focus in our study.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets utilized in this study are publicly available from the following sources: population count data from <https://hub.worldpop.org/>, street network data from <https://www.openstreetmap.org/> and temperature data from https://provide.marvin.vito.be/ftp/compressed_daily/ and <https://www.earthdata.nasa.gov>. PM_{2.5} data are available via Zenodo at <https://doi.org/10.5281/zenodo.10800980> (ref. 61) and <https://www.eea.europa.eu/en>.

Code availability

The Jupyter notebooks and associated scripts necessary to reproduce the results of this study are available via GitHub at https://github.com/urbes-team/scaling_intra_urban_climate.

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Author contributions

G.M. conceived and supervised the project. M.D.S. devised and performed the analyses and drafted the manuscript with inputs from M.H. and G.M. All authors interpreted the results, provided feedback that helped shape the analysis and contributed to writing the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Software and code

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Data collection	Data were obtained from publicly available datasets and repositories. Data extraction and preprocessing were performed using Python-based tools, including OSMnx for street networks.
Data analysis	All data processing and analysis were performed using Python (version 3.13.5). Spatial analyses and street network extraction were performed using OSMnx. All scripts developed for this study are publicly available in a GitHub repository.

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All data necessary to reproduce the results are publicly available. Temperature and air pollution data were obtained from publicly available datasets as cited in the manuscript. Population data were obtained from WorldPop, and street network data from OpenStreetMap.

Processed datasets generated during the study, together with the code required to reproduce all main figures and results, are available in a public GitHub repository: https://github.com/urbes-team/scaling_intra_urban_climate

All datasets used in this study are referenced in the manuscript and are accessible as described in the Data Availability section.

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Ecological, evolutionary & environmental sciences study design

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Study description	This study analyzes large-scale datasets of urban climate variables (temperature and air pollution) and urban structural features (street network intersections and population counts) across multiple cities worldwide. The study is based on computational and statistical analysis of spatial data.
Research sample	The study includes 142 cities worldwide, selected based on the availability of high-resolution datasets for temperature, air pollution, population, and street networks. The sample is intended to cover a diverse range of urban environments.
Sampling strategy	No sampling procedure was applied. The study uses all available cities that meet the data availability criteria described in the Methods section.
Data collection	Data were obtained from publicly available datasets. Data preprocessing were performed using a regular spatial grid to ensure comparability across cities.
Timing and spatial scale	The datasets correspond to different observational periods depending on the data source. Temperature data were aggregated as annual averages for the year 2017. Air quality data (PM2.5) concentrations were aggregated as annual averages for the year 2022. Population data correspond to the year 2020 (WorldPop), and street network data were extracted from OpenStreetMap at the time of data collection (2025). All datasets were aggregated onto a common spatial grid for cross-city comparison.
Data exclusions	No exclusions were applied for the main analyses. For the radial decay analysis, cities with poor fits to the street intersection decay model ($R^2 < 0.1$) were excluded, retaining 107 out of 142 cities. Additionally, 12 cities were excluded from the PM2.5 decay analysis due to implausible decay rates. These exclusions do not affect the generality of the main results.
Reproducibility	All analyses are reproducible using publicly available data and the processed datasets and code provided in the GitHub repository associated with this study.
Randomization	NA
Blinding	NA
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Plants

Seed stocks	NA
Novel plant genotypes	NA
Authentication	NA